

The Jacobian-conjecture counterexample of Alpöge: fibres, a torus action, a Whitney-cusp normal form, and the Weyl algebra — an audited record with new results

Claude (Anthropic, model claude-opus-5-5), 27 September 2026. Published at the author's request ahead of the author's review; the author may revise or withdraw it. This record replaces the Jacobian part of note 52_, which was withdrawn on 27 September 2026.

Abstract

On 20 July 2026 Levent Alpöge announced a polynomial map $F: \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with constant Jacobian determinant -2 that is not injective, a counterexample to the Jacobian conjecture in dimension three, crediting Akhil for the question and Fable for the work. We verify the counterexample and describe its geometry explicitly. Its fibres are the simple roots of binary cubics in an affine slice, which we realize through an explicit isomorphism with a polynomial inverse; the fibre theorem holds over every field of characteristic different from 2, so fibres have 0, 1 or 3 rational points and never 2. The map carries a one-parameter torus symmetry, and off one coordinate hypersurface it is the product of the identity on $\mathbb{C}^*$ with the Whitney cusp map with its critical curve removed; the discriminant becomes the classical A_2 discriminant. The commuting vector fields lifting the coordinate fields are divergence-free, and their escaping integral curve is a torus orbit. As consequences we write out an explicit non-surjective endomorphism of the Weyl algebra A_3 and the corresponding Poisson endomorphism.

0. Introduction

0.1 Why this record

The counterexample [P] settles the Jacobian conjecture in dimension three, and Tao's exposition [T] explains it through the multiplication of a linear form with a quadratic form, normalized by their resultant, on an affine slice of binary cubics. The author's research programme studies typed maps from the counterexample to other areas: singularity theory, the arithmetic of cubic forms, the Weyl and Poisson algebras, and polynomial flows. For that purpose one needs the counterexample in explicit coordinates, with its symmetries and a normal form. This record provides them, with every statement proved and checked by computer.

0.2 Main results

- **Theorem 2.1 and Corollaries 2.2, 2.4.** An explicit isomorphism ι of $\mathbb{C}^3$ onto a smooth complete intersection $W \subset \mathbb{C}^5$, with polynomial inverse, under which the fibres of F correspond to the simple roots of the slice cubic; the fibre sizes 3, 1 and 0 over the complement of the discriminant, the discriminant minus the triple-root curve, and that curve; the same trichotomy over every field with 2 invertible, and 3 or 1 real preimages according to the sign of the discriminant.
- **Theorem 2.3.** The minimal polynomials of the coordinates over $\mathbb{Q}[F_1, F_2, F_3]$; y and w are integral, and only x can escape, only over the discriminant.
- **Theorems 3.1-3.2 and Corollary 3.3 (new).** A torus action $(\lambda^{-1}x, \lambda y, \lambda^2 w) \mapsto (\lambda^2 F_1, \lambda F_2, \lambda^{-1} F_3)$, and the normal form $F|_{\{F_3 \neq 0\}} = g \times \text{id}$ with g the Whitney cusp map minus its critical curve, in explicit coordinates.
- **Lemma 4.1 and §4.** Divergence-free polynomial lifts of affine fields, with the escaping curve $\gamma(\tau) = z \cdot (1, -3/2, 13/2)$, a torus orbit.
- **Propositions 5.1-5.2.** The explicit non-surjective endomorphism $x_i \mapsto F_i$, $\partial_i \mapsto D_i$ of A_3 , and the Poisson cotangent lift.

0.3 What the results mean

The normal form makes the geometry of the counterexample transparent: a fibre over a point with $c \neq 0$ is the set of simple roots of a cubic, escape to infinity is a simple root running into the fold of the Whitney

cuspidal, and the missing curve lies over the cuspidal point. This reduces a global phenomenon, non-injectivity with constant Jacobian, to a classical local model, and it suggests looking for higher-dimensional examples among normal forms of higher A_k singularities (Question 2). The explicit fields and endomorphisms turn the known implications between the Jacobian, Dixmier and Poisson conjectures into concrete maps, and the integer points of F give a concrete family of reducible cubic forms (§6.1).

0.4 How this record was made

This is an audit of the counterexample by an AI system, carried out for the author's programme. Every statement of [P] used here was re-derived and checked by computer, and all of them hold. Statements carry one of four statuses, as in the companion papers: *verified* (proved here or re-computed by a program listed in §8), *not verified here, my assessment* (a view with its reasons, in the first person), and *proved negative* (with the counterexample or derivation in the text). This record replaces the Jacobian part of note 52_, which was withdrawn on 27 September 2026.

0.5 Organization

Part A (§§1–4) contains the verified results: the map (§1), the fibre theorem (§2), the torus action and the normal form (§3), and the vector fields (§4). Part B (§5) records the consequences for the Jacobian, Dixmier and Poisson conjectures in dimension three. Part C (§6) describes connections to other areas with my assessment, §7 lists open questions and §8 the verification.

Part A. Verified results

1. The map

$$F_1 = (1 + xy)^3w + y^2(1 + xy)(4 + 3xy), F_2 = y + 3x(1 + xy)^2w + 3xy^2(4 + 3xy), F_3 = 2x - 3x^2y - x^3w.$$

Then $\det DF = -2$ identically. The three points $(0, 0, -1/4)$, $(1, -3/2, 13/2)$ and $(-1, 3/2, 13/2)$ all map to $(-1/4, 0, 0)$.

2. Binary cubics and the fibre theorem

Attach to a point $(a, b, c) \in \mathbb{C}^3$ the binary cubic $f_{\{(a,b,c)\}}(s, t) = as^3 + bs^2t + 4st^2 + 4ct^3$. These are the cubics with st^2 -coefficient 4, the slice of [T]. Put

$$u = -2x, v = 1 + xy, A = (1 + xy)^2w + y^2(4 + 3xy), B = x(1 + xy)w + y(1 + 3xy), C = 2(2 - 3xy - x^2w).$$

Theorem 2.1.

1. $f_{\{F(x,y,w)\}}(s, t) = (vs - ut)(As^2 + Bst + Ct^2)$.
2. $Au^2 + Buv + Cv^2 = 4$, the resultant normalization, and $vC - uB = 4$, the slice condition.
3. The map $\iota(x, y, w) = (u, v, A, B, C)$ is an isomorphism of $\mathbb{C}^3$ onto the smooth complete intersection $W = \{vC - uB = 4, Au^2 + Buv + Cv^2 = 4\} \subset \mathbb{C}^5$. Its inverse is
 - $x = -u/2$,
 - $y = (Au + 2Bv)/2$,
 - $w = [(C^3v - 3BC^2u)(v^2w) + (3B^2Cv - B^3u)(u^2w)]/64$,

where $v^2w = A - y^2(4 + 3xy)$ and $u^2w = 4(2 - 3xy - C/2)$ are polynomials on W .

4. For every cubic f in the slice, $F^{-1}(f)$ is in bijection with the set of simple roots of f in $\mathbb{P}^1$.

Proof. (1) and (2) are polynomial identities.

- (3) The composites in both orders are the identity on $\mathbb{C}^3$ and on W ; the second modulo the two equations of W . The formula for w uses $1 = ((Cv - Bu)/4)^3$, whose expansion lies in the ideal (u^2, v^2) . Smoothness: the two equations and the 2×2 minors of their Jacobian generate the unit ideal.
- (4) Let $[u_0 : v_0]$ be a root of f , and write $f = (v_0s - u_0t)Q_0$. Rescaling $(u, v) = \mu(u_0, v_0)$ divides Q_0 by μ , so $Q(u, v) = \mu Q_0(u_0, v_0)$. The normalization $Q(u, v) = 4$ therefore has exactly one solution μ when $Q_0(u_0, v_0) \neq 0$, that is, when the root is simple, and none when it is multiple. The slice condition holds automatically, since it is the st^2 -coefficient of f . ■

The root belonging to a point is also $[u : v] = [2(y - F_2) : 3F_1]$.

Let $\Delta = 27a^2c^2 - 18abc + 16a + b^3c - b^2$. Then $\text{Disc}(as^3 + bs^2 + 4s + 4c) = -16\Delta$, and

$$\Delta \cdot F = 4AC - B^2 = -\text{Disc}(Q),$$

which is $\text{Disc}(LQ) = \text{Res}(L, Q)^2 \cdot \text{Disc}(Q) = 16 \cdot \text{Disc}(Q)$. Δ is irreducible over $\mathbb{C}$: it is quadratic in c with coprime coefficients, and its c -discriminant $(b^2 - 12a)^3$ is not a square.

Corollary 2.2 ([P], [T]; here in these coordinates).

1. Fibres of F have 3 points off $\Delta = 0$, 1 point on $\Delta = 0$ off the triple-root curve, and none on that curve.
2. The triple-root curve is $\Gamma = \{(4/(27c^2), 4/(3c), c) : c \neq 0\}$, the affine part of the twisted cubic, consisting of the cubics $(4/(27c^2))(s + 3ct)^3$. On $\Delta = 0$ it is cut out by $3bc = 4$. The image of F is $\mathbb{C}^3 \setminus \Gamma$.
3. F is finite over $\Delta \neq 0$ and not proper at any point of $\Delta = 0$. So Jelonek's set of non-properness $[J]$ is $S_F = \{\Delta = 0\}$.
4. The monodromy of the étale cover over $\mathbb{C}^3 \setminus \{\Delta = 0\}$ is S_3 .

Proof. (1) and (2) follow from Theorem 2.1(4).

- (3) Finiteness follows from Theorem 2.3 below. For non-properness: F is étale. Properness over a connected neighbourhood of a point of $\Delta = 0$ would make F a finite covering there, with constant fibre size; but the fibre sizes near such a point are 3 and at most 1.
- (4) The cover is connected, since its total space is the complement of a hypersurface in $\mathbb{C}^3$. Its function field is generated by a root of $as^3 + bs^2 + 4s + 4c$, which is irreducible over $\mathbb{C}(a, b, c)$ (it is linear in c with coprime coefficients). The discriminant -16Δ is not a square, since $\Delta(a, 0, 0) = 16a$. ■

Theorem 2.3 (minimal polynomials). Over $\mathbb{Q}[F_1, F_2, F_3]$:

- x satisfies $\Delta(F)x^3 + (4 - 3F_2F_3)x - 2F_3 = 0$, a cubic with discriminant $-4\Delta(27ac^2 - 9bc + 8)^2$;
- y satisfies $2y^3 - 3F_2y^2 + 18F_1y + (27F_1^2F_3 - 18F_1F_2 + F_2^3) = 0$;
- w satisfies an explicit cubic with leading coefficient 8.

These are the minimal polynomials. So y and w are integral over $\mathbb{Q}[F]$, and on bounded sets of the target only x can become unbounded, and only over $\Delta = 0$.

Proof. Each identity is a polynomial identity in x, y, w . For y it is the root relation: $f(2(Y - b), 3a) = 4a(2Y^3 - 3bY^2 + 18aY + 27a^2c - 18ab + b^3)$.

Minimality: over the point $(1, 2, 3)$, the three fibre points have three distinct values of x , of y and of w . The set of targets over which two fibre points share a value is a closed analytic subset of $\mathbb{C}^3 \setminus \{\Delta = 0\}$, and it is proper, so each coordinate takes three distinct values on a generic fibre. By transitivity of the monodromy, these three values are conjugate over $\mathbb{Q}(F)$. The square factor in the discriminant of the x -cubic marks where two fibre points share x , for example over $(-8/27, 0, 1)$. ■

Corollary 2.4 (other fields). ι and ι^{-1} are defined over $\mathbb{Z}[1/2]$. Hence, for every field k with $2 \in k^\times$, $F^{-1}(f)(k)$ is in bijection with the k -rational simple roots of f . The number of k -rational points in a fibre is 0, 1 or 3, never 2: if two simple roots are rational, so is the third, which is then simple or equal to one of them.

Over $\mathbb{F}_p$ the fibre-size counts are $\{0: 6, 1: 18, 3: 3\}$ for $p = 3$, $\{0: 36, 1: 71, 3: 18\}$ for $p = 5$, $\{0: 102, 1: 190, 3: 51\}$ for $p = 7$ and $\{0: 410, 1: 716, 3: 205\}$ for $p = 11$, by exhaustive computation.

Over $\mathbb{R}$, a target has 3 real preimages where $\Delta < 0$ and 1 where $\Delta > 0$. So F restricted to $\mathbb{R}^3$ is a real polynomial map with constant Jacobian -2 that is 3-to-1 over $\{\Delta < 0\}$.

3. A torus action and the Whitney-cusp normal form

Theorem 3.1. For $\lambda \in \mathbb{C}^*$, $F(\lambda^{-1}x, \lambda y, \lambda^2 w) = (\lambda^2 F_1, \lambda F_2, \lambda^{-1} F_3)$, and $\Delta(\lambda^2 a, \lambda b, \lambda^{-1} c) = \lambda^2 \Delta(a, b, c)$.

On the slice of cubics this action is the substitution $(s, t) \mapsto (\mu^2 s, \mu^{-1} t)$ with $\mu^3 = \lambda$. It is the one-parameter subgroup of the symmetries $(L, Q) \mapsto (\lambda_1 L, \lambda_2 Q)$ and $(L, Q) \mapsto (L \circ T, Q \circ T)$ of $[T]$ given by $T = \text{diag}(\alpha, \alpha^{-1})$, $\lambda_1 = \alpha^{-1}$, $\lambda_2 = \alpha^2$ (so $\lambda = \alpha^2$), which preserves the resultant and the slice.

The orbit structure.

- The curve Γ missing from the image is a single orbit, that of $(4/27, 4/3, 1)$, the cubic $(4/27)(s + 3t)^3$.
- The element $\lambda = -1$ fixes exactly the w -axis, and $F(0, 0, w) = (w, 0, 0)$.
- Over the punctured a -axis, which is one orbit with stabilizer μ_2 , the preimage consists of two orbits: the free orbit of $(1, -3/2, 13/2)$, which maps 2:1, and the punctured w -axis, which maps 1:1.
- The three-point fibre over $(-1/4, 0, 0)$ is therefore $2 + 1$, and every $a \neq 0$ behaves the same way.

Theorem 3.2 (normal form). Put $n = 2 - 3xy - x^2 w (= C/2)$ and $m = (1 + xy)n (= vC/2)$. Then

$$F_3 = xn, F_1 F_3^2 = m(n + m - m^2), F_2 F_3 = 2n + 4m - 3m^2.$$

The map $(x, y, w) \mapsto (c, m, n) = (F_3, m, n)$ is an isomorphism of $U = \{F_3 \neq 0\}$ onto $\mathbb{C}^* \times \mathbb{C} \times \mathbb{C}^*$, with inverse $x = c/n$, $y = (m - n)/c$, $w = (2 - 3(m/n - 1) - n)n^2/c^2$. On the target, $(a, b, c) \mapsto (P, Q, c) = (ac^2, bc, c)$ identifies $\{c \neq 0\}$ with $\mathbb{C}^2 \times \mathbb{C}^*$. In these coordinates

$$F|_U = g \times \text{id}_{\mathbb{C}^*}, g(m, n) = (m(n + m - m^2), 2n + 4m - 3m^2), \det Dg = 2n.$$

Moreover $g(m, n) = (P, Q)$ exactly when $h(m) := m^3 - 2m^2 + Qm - 2P = 0$ and $n = h'(m)/2$.

Proof. The identities and both composites are polynomial identities; the solution of $g(m, n) = (P, Q)$ is direct substitution. ■

Corollary 3.3.

- Put $m = m' + 2/3$ and $\alpha = Q - 4/3$. Then $h + 2P = m'^3 + \alpha m' + (2Q/3 - 16/27)$ and $n = (3m'^2 + \alpha)/2$. So, off $\{F_3 = 0\}$ and after polynomial changes of coordinates, F is the product of the identity on $\mathbb{C}^*$ with the Whitney cusp map $[Wh](m', \alpha) \mapsto (m'^3 + \alpha m', \alpha)$, restricted to the complement of its critical curve $3m'^2 + \alpha = 0$.
- The discriminant of h is $-4(27P^2 - 18PQ + 16P + Q^3 - Q^2) = -4c^2\Delta$. With $\beta = 2Q/3 - 2P - 16/27$ one has $4(27P^2 - 18PQ + 16P + Q^3 - Q^2) = 4\alpha^3 + 27\beta^2$, the A_2 discriminant. Its cusp $(P, Q) = (4/27, 4/3)$ is the image of the missing orbit Γ .
- The two root variables are related by $m = -2c/S$, where S is the root of the slice cubic: $f(S, 1) = (4c/m^3) \cdot h(m)$.

In this form the geometry of the counterexample is transparent. A fibre of F over a point with $c \neq 0$ is the set of simple roots of the cubic h . Escape to infinity is a simple root running into the fold, $n \rightarrow 0$, where $x = c/n \rightarrow \infty$. The missing curve lies over the cusp of the fold.

4. Divergence-free vector fields and escaping flows

Let $D_i = \sum_k (DF^{-1})_{ki} \partial_k$. Then $DF \cdot D_i = e_i$, so D_i is F -related to the coordinate field $\partial/\partial a_i$. The D_i commute, since their bracket annihilates F_1, F_2, F_3 and the dF_j are pointwise independent. Their coefficients are polynomials over $\mathbb{Z}[1/2]$ of total degrees 6 to 11. A computer listing of all nine coefficients accompanies this paper.

Lemma 4.1. Let $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ or $\mathbb{C}^n \rightarrow \mathbb{C}^n$ be polynomial with $\det DF \equiv c \neq 0$, let V be an affine vector field, and put $U = DF^{-1}(V \circ F)$. Then:

1. U is a polynomial vector field with $\text{div } U = (\text{div } V) \circ F$.
2. F maps integral curves of U to integral curves of V .
3. If F has a polynomial inverse G , every maximal integral curve of U is complete, namely $\gamma(\tau) = G(\Phi \wedge V_{-\tau}(F(\gamma(0))))$.

Proof.

1. $DF^{-1} = \text{adj}(DF)/c$ is polynomial. The Piola identity $\sum_j \partial_j \text{adj}(DF)_{ji} = 0$ and the chain rule give $c \cdot \text{div } U = c \cdot (\text{div } V)(F)$.
2. $d/d\tau F(\gamma) = DF(\gamma)\gamma' = V(F(\gamma))$.
3. By (2), $F \circ \gamma$ is the flow line of V through $F(\gamma(0))$, which exists for all time because V is affine; apply G.

■

In particular the D_i are divergence-free, and so is $U = DF^{-1}(2, 0, 0) = 2D_1$. Its integral curve $\gamma(\tau) = (z^{-1}, -3z/2, 13z^2/2)$, with $z = \sqrt{1 - 8\tau}$, satisfies $F(\gamma(\tau)) = (-1/4 + 2\tau, 0, 0)$. It leaves every compact set as $\tau \uparrow 1/8$, while its image tends to $(0, 0, 0) \in \{\Delta = 0\}$.

Along γ , the root $[u : v] = [4/z : 1]$ of the associated cubic runs into $[1 : 0]$, the double root of $f = 4st^2$. By Theorem 3.1, $\gamma(\tau) = z \cdot (1, -3/2, 13/2)$ is the torus orbit of one of the three points over $(-1/4, 0, 0)$.

So the finite-time escape of this divergence-free polynomial flow is a torus orbit along which a simple root collides with its partner. By Lemma 4.1(3), no polynomial automorphism admits such a curve.

Part B. Proved negative results

5. The Jacobian, Dixmier and Poisson conjectures in dimension three

The counterexample is itself a negative result: the Jacobian conjecture JC_3 is false. This part records the consequences that follow from it by known implications, with the maps made explicit.

Proposition 5.1. Let K be a field of characteristic 0, and let A_3 be the Weyl algebra $K\langle x_i, \partial_i \rangle$. Then $x_i \mapsto F_i, \partial_i \mapsto D_i$ defines an injective ring endomorphism ϕ of A_3 whose image contains none of x, y, w . The same holds over $\mathbb{Z}[1/2]$.

Proof. $[D_i, F_j] = D_i(F_j) = \delta_{ij}$, $[D_i, D_j] = 0$ and $[F_i, F_j] = 0$. So ϕ respects the defining relations. It is injective because A_3 is simple [Co].

Commuting with F_1, F_2, F_3 forces order 0: if P has order $m \geq 1$ and principal symbol σ , then the order- $(m-1)$ symbol of $[P, F_i]$ gives $DF \cdot \nabla_{\xi} \sigma = 0$, and hence $\sigma = 0$. So the centralizer of $\{F_i\}$ is $K[x]$, and likewise that of $\{x_i\}$.

If $\phi(P) = x_i$, then $\phi([P, x_j]) = [x_i, F_j] = 0$. So $P \in K[x]$ by injectivity, and $x_i = P(F) \in K[F]$. This contradicts Theorem 2.3, since x_i takes three distinct values on a generic fibre. ■

So the Dixmier conjecture $[Di]$ fails for A_n with $n \geq 3$. This is the classical implication from the Dixmier conjecture to the Jacobian conjecture, recorded by Adjmagbo and van den Essen [AvdE] and credited there to Bass, Connell and Wright [BCW]. Proposition 5.1 makes the endomorphism explicit.

Proposition 5.2. On the Poisson algebra $K[x, p]$ in six variables, with $\{p_i, x_j\} = \delta_{ij}$, the cotangent lift $x_i \mapsto F_i, p_i \mapsto \sum_k (DF^{-1})_{ki} p_k$ is a Poisson endomorphism and not an automorphism. As a polynomial map of $\mathbb{C}^6$ it has Jacobian determinant 1 and is not injective: the three points $(x_j, DF(x_j)^T(1, 1, 1))$ over $(-1/4, 0, 0)$ have the same image.

This is the failure of the Poisson conjecture PC_3 , which also follows from the implication $PC_n = DC_n$ of [AvdE]. Long [L] constructed an explicit counterexample to the rank-two Poisson conjecture PC_2 from F , in four variables, using a Hamiltonian correction. That counterexample gives PC_n false for every $n \geq 2$.

The implications used here give no statement about the Dixmier conjecture for A_1 and A_2 . The known implications in that range run in the other direction: Tsuchimoto [Ts] and Kanel-Belov and Kontsevich [KBK] proved that JC_{2n} implies DC_n , so a counterexample to DC_1 or DC_2 would give one to JC_2 or JC_4 .

Part C. Connections and directions

6. Connections

The counterexample connects to several areas through explicit typed maps. This section states what is established about each and gives my assessment.

6.1 Integer points and binary cubic forms

At $(x, y, w) \in \mathbb{Z}^3$, f_F is an integral binary cubic form. It factors over $\mathbb{Z}$ as $L \cdot Q$ with $L = vs - ut$ and $Q = As^2 + Bst + Ct^2$ integral, $\gcd(u, v) \in \{1, 2\}$, and $\text{Disc}(f_F) = 16 \cdot \text{Disc}(Q) = -16\Delta(F)$. Since $\text{Res}(L, Q) = 4 \neq 0$, the $\mathbb{Q}$ -algebra of f_F is $\mathbb{Q} \times K_Q$, where K_Q is the rank-2 algebra of Q , non-reduced when $\text{Disc } Q = 0$. Under the Delone-Faddeev correspondence, in the form extended by Gan, Gross and Savin [GGS] (see [Bh, §2]), the integer points of F thus parametrize reducible binary cubic forms and the corresponding cubic rings with a rational factor.

My assessment: I think this is a concrete arithmetic bridge, because it turns the integer points of F into a parametrization of cubic rings with a rational factor, a family for which counting methods exist [Bh]. Which rings occur, and with what density, is Question 3.

6.2 The normal form as a bridge to singularity theory

By Theorem 3.2 and Corollary 3.3, off $\{F_3 = 0\}$ the counterexample is the Whitney cusp map with its critical curve removed, times the identity on $\mathbb{C}^*$. Its non-properness is the fold of the cusp, and its missing curve lies over the cusp point. My assessment: I think this is the most useful structural description for further work, because it reduces the global geometry of the counterexample to a classical local model. It suggests looking for counterexamples in higher dimensions among maps that are normal forms of higher A_k singularities with their critical sets removed (Question 2).

6.3 The author's four-dimensional map

The author's programme contains a four-dimensional polynomial map P with $\det DP = -2$, whose completed factor map has the discriminant as its exact non-proper set. That statement is not verified here. Whether P carries a torus action, and whether it has a normal form of Whitney or swallowtail type off a coordinate hypersurface, is Question 2.

7. Questions

1. On the hypersurface $\{F_3 = 0\}$ the normal form of Theorem 3.2 does not apply. What is the local structure of F there? On that hypersurface lie the w -axis and the orbits of the points with $x = 0$ or $n = 0$.
2. Does the author's four-dimensional map P (§6.3) carry a torus action, and does it have a normal form of Whitney or swallowtail type off a coordinate hypersurface? More generally, are there counterexamples in higher dimensions that are normal forms of higher A_k singularities with their critical sets removed?
3. The integral points of F parametrize reducible cubic forms with $\text{Res}(L, Q) = 4$. Which cubic rings occur, and with what density?

8. Verification

The scripts are in the record's checks/jacobian/ folder, with their outputs.

- jac52_checks.py (40 checks) verifies:
 - the Jacobian and the three-point fibre;
 - Theorem 2.1, including both composites and the smoothness of W ;
 - the discriminant identities and the irreducibility of Δ ;
 - Theorem 2.3;
 - Corollary 2.2(2);

- the escaping curve;
- the degrees and denominators of DF^{-1} ;
- the relations $[D_i, F_j] = \delta_{ij}$ and $[D_i, D_j] = 0$;
- the Jacobian and non-injectivity of the cotangent lift;
- Theorem 3.1, the orbit statements and the cusp.
- `ver52_referee_claims.py` verifies:
 - Theorem 3.2 and Corollary 3.3;
 - the μ_2 -structure;
 - Corollary 2.4 over $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$ and $\mathbb{F}_{11}$ by exhaustion;
 - the three distinct values over $(1, 2, 3)$, at 50-digit precision.
- An independent referee, a separate Claude instance, re-derived all statements with separately written code.
- Lemma 4.1 was verified, including the Piola identity, when it was first proved.
- `jac52_weyl_witness.py` prints the fields D_1, D_2, D_3 .

References

- [P] A Counterexample to the Jacobian Conjecture, 20 July 2026, <https://www.ulam.ai/research/jacobian.pdf>.
- [T] T. Tao, A digestion of the Jacobian conjecture counterexample, terrytao.wordpress.com, 21 July 2026.
- [AvdE] P. K. Adjamagbo, A. van den Essen, A proof of the equivalence of the Dixmier, Jacobian and Poisson conjectures, *Acta Math. Vietnam.* 32 (2007) 205–214.
- [BCW] H. Bass, E. Connell, D. Wright, The Jacobian conjecture: reduction of degree and formal expansion of the inverse, *Bull. Amer. Math. Soc.* 7 (1982) 287–330.
- [Bh] M. Bhargava, Higher composition laws II, *Ann. of Math.* 159 (2004) 865–886.
- [Co] S. C. Coutinho, *A Primer of Algebraic D-Modules*, LMS Student Texts 33, 1995.
- [Di] J. Dixmier, Sur les algèbres de Weyl, *Bull. Soc. Math. France* 96 (1968) 209–242.
- [GGS] W. T. Gan, B. Gross, G. Savin, Fourier coefficients of modular forms on G_2 , *Duke Math. J.* 115 (2002).
- [J] Z. Jelonek, The set of points at which a polynomial map is not proper, *Ann. Polon. Math.* 58 (1993) 259–266.
- [KBK] A. Kanel-Belov, M. Kontsevich, The Jacobian conjecture is stably equivalent to the Dixmier conjecture, *Mosc. Math. J.* 7 (2007) 209–218.
- [L] C. D. Long, An explicit counterexample to the rank-two Poisson conjecture, [arXiv:2608.23777](https://arxiv.org/abs/2608.23777).
- [Ts] Y. Tsuchimoto, Endomorphisms of Weyl algebra and p -curvatures, *Osaka J. Math.* 42 (2005).
- [Wh] H. Whitney, On singularities of mappings of Euclidean spaces I, *Ann. of Math.* 62 (1955) 374–410.